List 4 - Fin. Ele

1. Let $f \in C([0,1])$ be given. For real-valued, piecewise continuous and bounded functions on $[0,1]$ define

$$(v,w) = \int_0^1 v(x)\,w(x)\,dx,$$

Let

$$V = \left\{ v \in C([0,1]) \;:\; v' \text{ is piecewise continuous and bounded, } v(0) = v(1) = 0 \right\},$$

and define $a(v,w) := (v',w')$ and $F : V \to \mathbb{R}$ by $F(v) = \frac{1}{2}a(v,v) - (f,v)$.

Consider the three problems

(DF) Find $u \in C^2([0,1])$ such that $-u''(x) = f(x)$ for $0 < x < 1$, $u(0) = u(1) = 0$;

(VF) Find $u \in V$ such that $a(u,v) = (f,v)$ $\forall v \in V$;

(EF) Find $u \in V$ such that $F(u) \le F(v)$ $\forall v \in V$.

Prove that the problems **(DF)**, **(VF)** and **(EF)** are equivalent, and that each of them has exactly one solution.

(DF) => (VF)

Let u be a solution. $u \in V$, because $u(0)=u(1)=0$ and $u \in C^2([0,1])$

Take $v \in V$. Then:

$$-u''v = fv \Rightarrow -\int_0^1 u''v\,dx = \int_0^1 fv\,dx \qquad (u'v)' = u''v + u'v' \Rightarrow$$

$$\int_0^1 fv\,dx = -\int_0^1 u'v'\,dx + \int_0^1 u'v'\,dx = u'v\Big|_0^1 + \int_0^1 u'v'\,dx$$

$$\Rightarrow a(u,v) = (f,v)$$

(VF) => (DF)

u solution. $u \in V$,

$$\int_0^1 u'(x)v'(x)\,dx = \int_0^1 f(x)v(x)\,dx, \quad \forall v \in V \Rightarrow$$

$$\Rightarrow -\int_0^1 u''(x)v(x)\,dx = \int_0^1 f(x)v(x)\,dx \Rightarrow$$

$$\Rightarrow -\int_0^1 (u''(x)+f(x))v(x)\,dx = 0, \quad \forall v \in V$$

Supposing $u'' \in C^0$: suppose by absurd that $u''+f \not\equiv 0$. $\exists x_0 \in [0,1]$ such that $u''(x_0)+f(x_0) > 0$. By the intermediate value theorem, $\exists \varepsilon > 0$ s.t. if $x \in B_\varepsilon(x_0)$, $\ddot{u}(x)+f(x) > 0$. Take v s.t. $v(x) > 0$ if $x \in B_\varepsilon(x_0)$ and $v(x) = 0$ if $x \notin B_\varepsilon(x_0)$

Then $$\int_0^1 (u''(x)+f(x))v(x)\,dx = \int_{x_0-\varepsilon}^{x_0+\varepsilon} (\ddot{u}(x)+f(x))v(x)\,dx > 0$$

absurd. So $-u'' = f$

VF => EF

Let u solution of VF, $v \in V$. $w := v - u \in V$

$$F(v) = F(u+w) = \frac{1}{2}\langle u'+w', u'+w'\rangle - \langle f, u+w\rangle =$$

$$= \frac{1}{2}\left(\langle u', u'+w'\rangle + \langle w', u'+w'\rangle\right) - \langle f,u\rangle - \langle f,w\rangle =$$

$$= \frac{1}{2}\left(\langle u',u'\rangle + \langle u',w'\rangle + \langle u',w'\rangle + \langle w',w'\rangle\right) - \langle f,u\rangle - \langle f,w\rangle$$

$$= F(u) + \frac{1}{2}\langle w',w'\rangle + \underbrace{\langle u',w'\rangle - \langle f,w\rangle}_{0} = F(u) + \frac{1}{2}\|w'\|^2$$

$$\Rightarrow F(v) = F(u) + \frac{1}{2}\|w'\|^2 \Rightarrow F(u) \ge F(v)$$

EF => VF

u solution. $v \in V$, $\varepsilon > 0$

$$F(u+\varepsilon v) = \frac{1}{2}\langle u'+\varepsilon v', u'+\varepsilon v'\rangle - \langle f, u+\varepsilon v\rangle =$$

$$= \frac{1}{2}\langle u',u'\rangle + \frac{\varepsilon^2}{2}\langle v',v'\rangle + \varepsilon\langle u',v'\rangle - \langle f,u\rangle - \varepsilon\langle f,v\rangle$$

$h(\varepsilon) = F(u+\varepsilon v)$

As u is minimum, $h'(0) = 0$

$h'(\varepsilon) = \varepsilon\langle v',v'\rangle + \langle u',v'\rangle - \langle f,v\rangle$

$h'(0) = \langle u',v'\rangle - \langle f,v\rangle = 0 \Rightarrow \langle u',v'\rangle = \langle f,v\rangle \Rightarrow a(u,v) = \langle f,v\rangle$

Uniqueness

DF

Let u,w solutions. Define $z = u-w$

In particular

We have $z'' = 0$ and $z(0) = z(1) = 0$

integrating twice gives $z(x) = ax+b$

$z(0) = 0 \Rightarrow b = 0$ $z(1) = 0 \Rightarrow a = 0 \Rightarrow z \equiv 0 \Rightarrow$

$\Rightarrow u - w = 0 \Rightarrow u = w$, $\forall x \in [0,1]$

VF

u,w solutions

$a(u,v) = (f,v) \Rightarrow (u',v') - (w',v') = 0$

$a(w,v) = (f,v)$

$\Rightarrow (v', u'-w') = 0$. Def. $z = u-w \Rightarrow z(0) = z(1) = 0$

$\Rightarrow (v', z') = 0$

$$\Rightarrow \int_0^1 v'(x)z'(x)\,dx = 0, \quad \forall v \in V$$

As z' is piecewise continuous, we use the same argument as before $\Rightarrow z' = 0 \Rightarrow z(x) = C$

$z(0) = 0 \Rightarrow C = 0 \Rightarrow z(x) = 0 = u-w \Rightarrow u = w$

EF

u,w solutions

$F(u) = F(w) \Rightarrow$

$$\Rightarrow \frac{1}{2}a(u,u) - (f,u) = \frac{1}{2}a(w,w) - (f,w) \Rightarrow$$

$$\Rightarrow \frac{1}{2}\left((u',u') - (w',w')\right) = (f, u-w)$$

$$\Rightarrow \int_0^1 (u')^2 - (w')^2\,dx = 2\int_0^1 f(u-w)\,dx$$

$$= \int_0^1 (u')^2\,dx - \int_0^1 (w')^2\,dx$$

$(u'u)' = (u')^2 + u''u$

$$\int_0^1 (u')^2\,dx = u'u\Big|_0^1 - \int_0^1 u''u\,dx = -\int_0^1 u''u\,dx$$

$$\Rightarrow \int_0^1 w''w - u''u\,dx = 2\int_0^1 f(u-w)\,dx$$

They satisfy VF $\Rightarrow u = w$. End.

4. Write the variational formulation of the following problem that involves **Robin**[10] boundary conditions:

$$\begin{cases} -\nabla\cdot(\mu\nabla u) = f & \text{in } \Omega \\ u = g & \text{on } \Gamma_D \\ \mu\nabla u\cdot\check{\mathbf{n}} + \gamma u = h & \text{on } \Gamma_R \end{cases}$$

where $\gamma \in \mathbb{R}$ is a given parameter and $\Gamma_D \cup \Gamma_R = \partial\Omega$ (and $\Gamma_D \cap \Gamma_R = \emptyset$). In fluid mechanics, this condition serves for instance to model heat exchange between a solid wall and the ambient fluid by convection.

$$\int_\Omega \mu\nabla u\cdot\nabla v\,dx - \int_{\Gamma_D\cup\Gamma_R} \mu\nabla u\cdot\check{n}\,v\,ds = \int_\Omega fv\,dx$$

$$\int_{\Gamma_D\cup\Gamma_R} \mu\nabla u\cdot\check{n}v\,ds = \int_{\Gamma_R} \mu\nabla u\cdot\check{n}v\,ds = \int_{\Gamma_R}(h-\gamma u)v\,ds$$

$$\Rightarrow \int_\Omega \mu\nabla(u_0+u_g)\cdot\nabla v\,dx = \int_\Omega fv\,dx + \int_{\Gamma_R}(h-\gamma u)v\,ds$$

$$\Rightarrow \int_\Omega \mu\nabla u_0\cdot\nabla v\,dx = \int_\Omega fv\,dx + \int_{\Gamma_R}(h-\gamma u)v\,ds - \int_\Omega \mu\nabla u_g\cdot\nabla v\,dx$$

$$\int_\Omega \mu\nabla u_0\cdot\nabla v\,dx + \int_{\Gamma_R} \gamma u_0 v + \gamma u_g v\,ds = \int_\Omega fv\,dx - \int_\Omega \mu\nabla u_g\cdot\nabla v\,dx$$

Find $u_0 \in H^1_{D0}(\Omega)$ such that, $\forall v \in H^1_{D0}(\Omega)$

3. Consider the two-point boundary value problem

$$-(\kappa u')' = f, \quad 0<x<1, \qquad u(0)=0, \quad \kappa(1)u'(1)+\gamma u(1) = g,$$

where $g \in \mathbb{R}$, $\gamma > 0$, $f \in L^2(0,1)$, and $\kappa \in C^1([0,1])$ satisfies $\kappa(x) \geq \kappa_0 > 0$ on $[0,1]$. Let

$$\widetilde{V} = \{ v \in H^1(0,1) : v(0) = 0 \}, \qquad \ell(v) = (f,v) + g\,v(1),$$

$$a(u,v) = (\kappa u', v') + \gamma\, u(1)\, v(1).$$

a) Show that problem admits the variational formulation: find $u \in \widetilde{V}$ such that $a(u,v) = \ell(v)$ for all $v \in \widetilde{V}$.
b) Explain why the condition $u(0) = 0$ must be imposed through the space $\widetilde{V}$, whereas the condition at $x = 1$ need not.
c) Prove the one-dimensional inequality

$$|v(1)| \leq \sqrt{2}\|v\|_{H^1(0,1)} \qquad \forall v \in H^1(0,1).$$

d) Show that $a(\cdot,\cdot)$ is continuous and strongly coercive on $\widetilde{V}$, and that $\ell(\cdot)$ is continuous. Then, conclude that the problem is well posed, and exhibit the constants N_a, α and N_ℓ.
e) Since $a(\cdot,\cdot)$ is symmetric, state the energy functional J for this problem.

a) Let $v \in \widetilde{V}$

$$-(\kappa u')' = f \Rightarrow \int_0^1 -(\kappa u')'v\,dx = \int_0^1 fv\,dx$$

Notice that $(\kappa u'v)' = (\kappa u')'v + \kappa u'v'$. Substituting:

$$\Rightarrow (\kappa u')'v = (\kappa u'v)' - \kappa u'v'$$

$$\Rightarrow \int_0^1 \kappa u'v'\,dx - \int_0^1 (\kappa u'v)'\,dx = \int_0^1 fv\,dx \Rightarrow$$

$$\Rightarrow \int_0^1 \kappa u'v'\,dx - \kappa u'v\Big|_0^1 = \int_0^1 fv\,dx \Rightarrow$$

$$\Rightarrow \int_0^1 \kappa u'v'\,dx + \underbrace{\kappa(0)u'(0)}_{0}v(0) - \kappa(1)u'(1)v(1) = \int_0^1 fv\,dx$$

Using $\kappa(1)u'(1) = g - \gamma u(1)$

$$\int_0^1 \kappa u'v'\,dx = \int_0^1 fv\,dx + v(1)\big(g - \gamma u(1)\big)$$

$$\Rightarrow \int_0^1 \kappa u'v'\,dx + \gamma u(1)v(1) = \int_0^1 fv\,dx + v(1)g \Rightarrow a(u,v) = l(v)$$

b) At $x=0$ we have a Dirichlet BC. Note that the term related to this boundary is $\kappa(0)u'(0)v(0)$ and the value of u does not appear in it. This means that, if we don't set $v(0)=0$, we have a flux term in the BC that shouldn't appear. On the other hand, at $x=1$, the flux is defined, so we don't have this problem.

c) $\|v\|_{H^1} = \left(\int_0^1 v^2 + (v')^2\,dx\right)^{\frac{1}{2}}$

$$\int_0^1 v^2 + (v')^2\,dx = \int_0^1 v^2\,dx + \int_0^1 (v')^2\,dx = \int_0^1 v^2\,dx + v'v\Big|_0^1 - \int_0^1 v''v\,dx =$$

$$= \int_0^1 v^2\,dx + v'(1)v(1) - \underbrace{v'(0)}_{0}v(0) - \int_0^1 v''v\,dx =$$

$$= \int_0^1 v(v-v'')\,dx + v'(1)v(1) \geq -\int_0^1 vv''\,dx + v'(1)v(1)$$

$$v^2(1) - v^2(x) = \int_x^1 (v^2)'\,dt = \int_x^1 2vv'\,dt$$

$$\Rightarrow v^2(1) = v^2(x) + 2\int_x^1 2vv'\,dt$$

$$\Rightarrow \int_0^1 |v^2(1)|\,dx = |v^2(1)| = \int_0^1 |v^2(x)|\,dx + 2\int_0^1\left|\int_x^1 vv'\,dt\right|dx =$$

$$= \|v\|_{L^2}^2 + 2\int_0^1\left|\int_x^1 vv'\,dt\right|dx \leq \|v\|_{L^2}^2 + 2\int_0^1\int_x^1 |vv'|\,dt\,dx \leq$$

$$\leq \|v\|_{L^2}^2 + 2\int_0^1 |v||v'|\,dt \overset{CS}{\leq} \|v\|_{L^2}^2 + 2\left(\int_0^1 |v|^2\,dt\right)^{\frac{1}{2}}\left(\int_0^1 |v'|^2\,dt\right)^{\frac{1}{2}} =$$

$$= \|v\|_{L^2}^2 + 2\,\|v\|_{L^2}\|v'\|_{L^2} \overset{2ab\leq a^2+b^2}{\leq} \|v\|_{L^2}^2 + \|v\|_{L^2}^2 + \|v'\|_{L^2}^2 \leq$$

$$2\left(\|v\|_{L^2}^2 + \|v'\|_{L^2}^2\right) = 2\|v\|_{H^1}^2$$

$$\Rightarrow |v(1)| \leq \sqrt{2}\,\|v\|_{H^1}$$

d) $a(u,v) = (\kappa u', v') + \gamma u(1)v(1)$

Continuity:

$$a(u,v) = \int_0^1 \kappa u'v'\,dx + \gamma u(1)v(1)$$

$$|a(u,v)| \leq \left|\int_0^1 \kappa u'v'\,dx\right| + |\gamma||u(1)||v(1)| \leq$$

$$\leq \left|\int_0^1 \kappa u'v'\,dx\right| + 2|\gamma|\,\|u\|_{H^1}\|v\|_{H^1}$$

Let $\kappa_M = \max_{x\in[0,1]}|\kappa|$

$$\left|\int_0^1 \kappa u'v'\,dx\right| \leq \int_0^1 |\kappa u'v'|\,dx \leq \kappa_M\int_0^1 |u'v'|\,dx \leq \kappa_M\left(\int_0^1 (u')^2\,dx\right)\left(\int_0^1 (v')^2\,dx\right)$$

$$\leq \kappa_M\left(\int_0^1 u^2 + (u')^2\,dx\right)\left(\int_0^1 v^2 + (v')^2\,dx\right) = \kappa_M\,\|u\|_{H^1}\|v\|_{H^1}$$

$$\Rightarrow |a(u,v)| \leq \underbrace{(\kappa_M + 2\gamma)}_{N_a}\|u\|_{H^1}\|v\|_{H^1} \Rightarrow a(\cdot,\cdot) \text{ is continuous}$$

Coercivity:

$$a(u,u) = (\kappa u', u') + \gamma u^2(1) = \int_0^1 \kappa(u')^2\,dx + \gamma u^2(1) \geq \int_0^1 \kappa(u')^2\,dx \geq$$

$$\geq \kappa_0\int_0^1 (u')^2\,dx \geq \frac{\kappa_0}{2}C_{(0,1)\{0,1\}}\int_0^1 u^2\,dx + \frac{\kappa_0}{2}\int_0^1 (u')^2\,dx \geq$$

$$\geq \underbrace{\frac{1}{2}\min\left\{\kappa_0 C_{(0,1)\{0,1\}}, \kappa_0\right\}}_{\alpha}\|u\|_{H^1}$$

Continuity of l:

$$l(v) = (f,v) + g\,v(1) = \int_0^1 f v\,dx + g\,v(1)$$

$$|l(v)| \le \left|\int_0^1 f v\,dx\right| + |g|\,|v(1)| \le \int_0^1 |f v|\,dx + \sqrt{2}|g|\,\|v\|_{H^1}$$

$$\int_0^1 |f v|\,dx \le \left(\int_0^1 f^2\,dx\right)\left(\int_0^1 v^2\,dx\right) \le \left(\int_0^1 f^2\,dx\right)^{1/2}\left(\int_0^1 v^2 + (v')^2\,dx\right)^{1/2}$$

$$= \|f\|_{L^2}\,\|v\|_{H^1}$$

$$\Rightarrow |l(v)| \le \underbrace{\left(\|f\|_{L^2} + \sqrt{2}|g|\right)}_{N_l}\|v\|_{H^1}$$

e) $J(u) = \frac{1}{2}a(u,u) - l(u)$

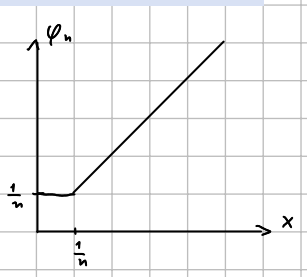
5. Let $\Omega = (0,1)$. Let $\varphi(x) = x$. Think on a sequence $\{\varphi_n\} \subset V_{N0}(\Omega) = \{v \in H^1(\Omega),\ v'(0) = 0\}$. i.e, $\varphi_n'(0) = 0$ for all n and such that $\varphi_n \to \varphi$ in the $H^1(\Omega)$-norm.
Hint: For $1/n = \epsilon > 0$ consider the "trimmed" function

$$T_{\epsilon(n)}\varphi(x) = \begin{cases} \varphi(\epsilon) & \text{if } x < \epsilon(n) \\ \varphi(x) & \text{if } x \ge \epsilon(n) \end{cases}$$

This example shows that the set of functions that satisfy a certain Neumann boundary condition does not necessarily contains all its accumulation points (in this case, the $V_{N0}(\Omega)$ is not closed in $H^1(\Omega)$).

$\varphi(x) = x$

$$\varphi_n(x) = \begin{cases} 1/n, & [0, 1/n] \\ x, & [1/n, 1] \end{cases}$$



$\varphi_n \to \varphi$

$e_n = \|\varphi_n - \varphi\|_{H^1} \to 0$

Abrir as contas e tender $n \to \infty$